Table S1. Species and data coverage

Taxon	Assembly / annotation	Data evaluated	Test scope
Arabidopsis thaliana	TAIR10/Araport11; Ensembl Plants release 62	Full annotation, chromosome FASTA, selected expression and protein examples	Full recognized-feature tuple audits; promoter, CDS QC, homology, expression arithmetic, domain mapping; app regressions
Oryza sativa	IRGSP-1.0; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Zea mays	B73 NAM5; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Glycine max	v2.1; Ensembl Plants release 62	Full annotation and source excerpt	Full tuple audit; multi-chromosome parser comparison
Vigna umbellata	GCF_029865625.1, Vumbellata_1.0; GCF_029865625.1-RS_2026_09	Full GFF3 and GTF; matching chromosome FASTA and RNA FASTA; three public read libraries	Full imports; query/structure; independent selected-sequence gold; Salmon re-quantification and exact ID reconciliation
Andrographis paniculata	GCF_009805555.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction
Hibiscus syriacus	GCF_006381635.1; Annotation Release 100	Full GFF3; selected scaffold FASTA	Full GFF3 import/search; selected-locus structure and sequence reconstruction